

The Strategic Value of AI-Driven Physical Identity and Access Management: Establishing Cyber-Physical Zero Trust in Critical Infrastructure

¹Prudhvi Ananthula

Masters in computer science
University of Michigan
prudhvia16@gmail.com
pananthularesearch@gmail.com
ORCID: 0009-0003-5458-8989
North Brunswick, New Jersey, USA

³Mayank Sethi

Bachelor's of Technology
Independent Researcher
mayanksethi86@gmail.com
ORCID ID: 0009-0006-6853-0878
IEEE Membership No: 101499983
Austin, Texas, USA

²Shruthi Sepuri

Masters in Applied Computer Science
Tata Consultancy Services Limited
shruthi.ss0307@gmail.com
ORCID ID: 0009-0000-6573-9752
Schaumburg, IL, USA

⁴Himanshu Goel

Independent Researcher
himanshu997goel@gmail.com
ORCID ID: 0009-0007-3217-176X
Delhi, India

Abstract—The rapid convergence of physical and digital systems has transformed critical infrastructure into highly interconnected cyber-physical environments. While this integration improves operational efficiency, it also introduces complex security challenges, as traditional perimeter-based security models fail to address threats that exploit both physical and cyber domains. Existing Physical Identity and Access Management (PIAM) systems primarily rely on static authentication mechanisms such as access cards and biometrics, which lack real-time adaptability and are vulnerable to credential theft, spoofing, and insider threats. Recent approaches have attempted to incorporate Zero Trust principles and advanced authentication techniques; however, they often remain limited to cybersecurity layers and fail to fully integrate physical access control with continuous verification mechanisms. As a result, these systems are unable to provide holistic protection across cyber-physical infrastructure. To address these limitations, this paper proposes an AI-driven PIAM framework that establishes a cyber-physical Zero Trust architecture. The proposed system integrates biometric authentication, behavioural analytics, and context-aware access control using machine learning models to enable continuous verification of users. Experimental results demonstrate that the proposed approach achieves an accuracy of 97%, significantly improving threat detection capability and reducing unauthorized access risks through dynamic and intelligent decision-making.

Keywords—Artificial Intelligence, Physical Identity and Access Management (PIAM), Zero Trust, Cyber-Physical Security, Critical Infrastructure, Behavioural Analytics, Access Control

I. INTRODUCTION

The rapid advancement of digital technologies has significantly transformed the operation of critical infrastructure systems [2]. Sectors such as energy,

transportation, healthcare, and industrial manufacturing are increasingly adopting interconnected systems that include physical processes with digital control mechanisms. These cyber-physical systems enhance efficiency, automation, and real-time monitoring [3]. However, this integration has enhanced the attack surface as well as putting infrastructure at risk of complex security threats that cut across physical and cyber space [11].

In traditional systems, the security within an environment is mostly based on the model of perimeter, in which trust is granted to the user and the devices within a line of demarcation. Entities tend to have wide access privileges without verification after authentication. Even the previous systems cannot be applied to the current distributed environments [12]. Insider attacks, stealing of credentials, and unauthorised access are some of the threats that can attempt to circumvent these defences and result in severe consequences, like failure of systems, data breaches, and safety.

Physical Identity and Access Management (PIAM) are systems that manage access to physical areas in critical infrastructure [5]. These systems normally require keycards, PINs or biometric authentication that provides security, is largely inert and is not able to change with changing times [7]. After gaining access, the user behaviour is not monitored or reassessed continuously. This restricts them, particularly where there are stolen credentials or where there is insider abuse [3]. Meanwhile, the Zero Trust model is a successful security method in cybersecurity. It works on the rule, like never trust, always verify, to ensure access requests can be sorted out [6]. The key aspects of this model are least-privilege access, real-

time monitoring, and making decisions regarding the context [10]. Although Zero Trust has proved effective in digital systems, it is limitedly used in physical security [1]. This integration failure of physical and cyber security provides a major gap in the protection of modern infrastructure.

The recent studies show that AI is most efficient at improving security systems [5]. Machine learning systems are able to process the access logs, sensors, and surveillance systems in large amounts of data and identify abnormal patterns and forecast danger [4]. The AI in PIAM systems can be used to implement dynamic and context-aware access control, based on user behaviour, location and time factors. This makes it possible to make decisions that are smarter and flexible [9].

Even with these advancements, most of the existing solutions address physical and cyber security individually and do not propose a unified framework [15]. They are usually not scalable or flexible and fail to respond to threats dynamically [3]. To address these limitations, the following paper proposed an AI-based PIAM system that creates the cyber-physical zero-trust architecture [8]. Multifactor authentication incorporates behavioural analytics and context-aware access control, and biometric authentication is integrated within the system to provide constant verification [14].

The approach proposed by the system increases the protection by detecting threats in real-time, making decisions based on access, and achieving greater resilience in the system [15]. It minimises the risks of insider attacks and unauthorised access and keeps the operations more efficient [5]. Overall, the given work presents a scalable and smart solution that effectively addresses the gap between physical and cyber security in critical infrastructure systems [6].

II. RELATED WORKS

The latest developments in the field of cybersecurity research point to the growing necessity of dynamic and intelligent security systems in order to defend critical infrastructure [7]. Conventional perimeter-based security models are less applicable in the contemporary environment of cyber-physical security threats, in that they cannot accommodate dynamic and multi-dimensional threats [8]. These models are based on predetermined limits that are easily violated through advanced attacks. In order to overcome these constraints, scholars are working on the combination of the Zero Trust concept with artificial intelligence (AI) in order to enhance the level of security and the effectiveness of decision-making [3].

Zero Trust architecture has become a significant paradigm of security that eliminates implicit trust and establishes the policy of constant verification of users and devices [8]. Zero Trust does not presume that objects in a network are trustworthy but demands that each request to access be authenticated according to numerous criteria. Investigations have also shown that it works well in the cloud and enterprise world by limiting unauthorised access and insider threats [10]. However, the adoption of Zero Trust presents a number of challenges, such as scalability, having system-to-system interoperability, and complexity in enforcing policies [6]. Furthermore, the majority of the current strategies mainly concentrate on electronic systems and fail to provide sufficient control over the physical access.

Zero Trust architectures based on AI have been suggested to manage new and sophisticated threats in the critical infrastructure domains like healthcare and power systems [11]. The solutions add protection to the network and application layers through intelligent threat detection and response. They, however, tend to ignore the role of Physical Identity and Access Management (PIAM). The old system of PIAM is based on access cards, PINs, and biometric authentication systems that offer simple security, but no real-time monitoring and flexibility [15]. It has been shown that behavioural analysis can be effectively used to enhance the identification of abnormal user behaviour and possible threats [14].

Anomaly detection and predictive analysis are capabilities enhanced by the implementation of AI in security systems, mostly in the fields of IoT and industry [15]. Context-aware access control systems, too, play their role, as they allow the use of dynamic decision-making ratified by the factors of user behaviour, location, and the level of risk [12]. Regardless of these advances, there are a lot of systems that are not aligned with the Zero Trust principles and are usually independent ones.

Overall, there is still a serious void in the creation of an integrated system that incorporates AI, PIAM, and Zero Trust [12]. The current solutions consider cyber and physical security as distinct domains, which makes them have fragmented protection [9]. Hence, it is apparent that the development of a detailed cyber-physical security model that will guarantee a high stability of protection of critical infrastructure systems is necessary [13].

III. PROPOSED FRAMEWORK

The proposed system introduces an AI-driven Physical Identity and Access Management (PIAM) framework designed to establish a cyber-physical Zero Trust security model for critical infrastructure. The main objective of this

system is the removal of an implicit trust both in the physical and online access environments and the replacement of this trust with the unceasing, context-sensitive verification. Unlike the traditional access control systems that utilise the static authentication methods, the proposed system contains artificial intelligence to facilitate dynamic decision-making and adjustable security policies.

The need for such a system is due to the limitations of existing PIAM and Zero Trust implementations. Conventional PIAM systems mainly focus on one-time identity verification using credentials such as access cards or biometrics. After the access is granted, the user behaviour is not monitored constantly. Similarly, in many Zero Trust systems, physical access control is omitted, with physical access control implemented only at the network or application interface. This separation creates a serious security gap in the physical environments of cyber-physical space, where threats may emerge in both spaces. The proposed system overcomes this gap by integrating physical and cyber security into one smart system. The main idea of the proposed framework is that it integrates identity checks with continuous behaviour and context checking and applies Zero Trust to physical access control.

The proposed system has an architectural design consisting of several interconnected layers, which collaborate to guarantee access control security. The layers are identity verification, behavioural analysis, context-sensitive decision-making and continuous authentication. The layers are improved with artificial intelligence to increase the accuracy, versatility, and responsiveness.

The identity verification application involves the use of advanced technology for verifying identity through facial recognition and fingerprint scanners. These approaches are chosen because of their uniqueness and inability to be imitated in comparison to traditional credentials. However, biometric systems cannot be considered to be enough because they may be subject to spoofing or abuse. Thus, the proposed system complement biometric verification with AI-based validation methods that may identify anomalies like facial mismatch or abnormal authentication history. The overall plan of the proposed AI-based PIAM system is shown in Fig. 1. It demonstrates the combination of identity verification, behavioural analysis, contextual evaluation, risk evaluation, and constant monitoring in an integrated cyber-physical zero-trust system.

The behavioural analysis engine is another important part of the system that uses machine learning algorithms to model normal user behaviour. This contains patterns like access frequency, movement within the facilities, time of entry, and interaction with the systems. This can be achieved especially with the unsupervised learning

methods, including clustering and anomaly detector algorithms, which can detect a violation of pre-existing behaviour patterns without labelled information. Through constant learning on the past data, the system is able to identify possible suspicious activities in real time and cause the necessary security measures to be taken.

Figure 1: Block diagram of the AI-driven PIAM Framework, detailing the integration of cyber-physical infrastructure inputs, core identity and continuous monitoring operational layers, and the underlying AI & Machine Learning engine for critical infrastructure security.

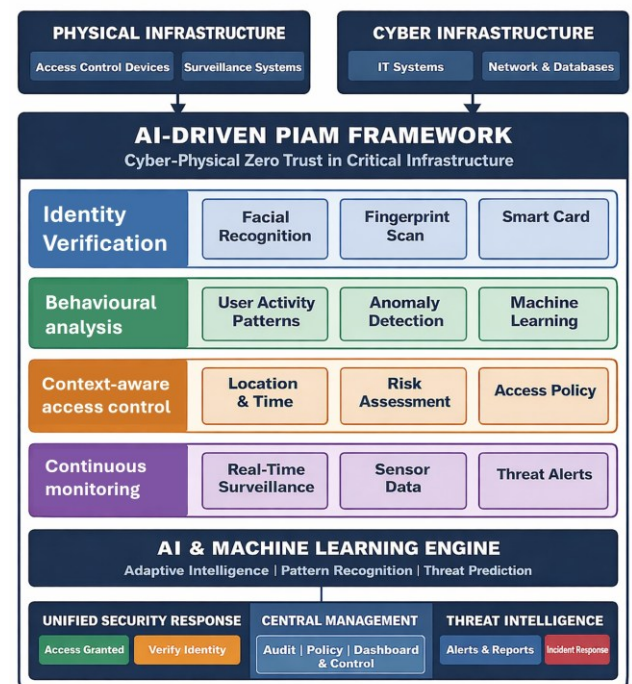


Fig. 1. Overall Architecture of the Proposed AI-Driven PIAM Framework

Besides a behavioural analysis, a context-aware access control mechanism is also included in the system. Identity is not the only parameter used in access decisions but also the context of the parameters like location, time, device status, and environmental parameters. The system does not allow the user in strange times like late night or from unknown places, which could be treated as high-risk. It is possible to evaluate these contextual factors with the help of machine learning models, such as decision trees and ensemble methods, and provide a risk score to every access request. Such a risk-based strategy is dynamic in controlling access policies as well as ensuring that policies are responsive to evolving situations.

The idea of continuous authentication is the main one of the proposed systems. The proposed framework, in comparison to the traditional systems, is able to check the identity of

users only when they enter the system and not during the entire session. This is done by combining information from various sources such as surveillance systems, sensors and access logs. The AI models process this information and make sure that the behaviour of the user is in line with their profile. In case of any variation, the system is capable of automatically negating access or implementing a further procedure.

The other main element of the proposed system is the combination of physical and cyber security. The system integrates physical access control systems like doors, biotic scanners, and monitoring cameras with IT systems like databases, networks and monitoring platforms. This integration allows viewing all the security events on both domains so that more extensive threats can be detected. The choice of AI as the main enabling technology is conditioned by the possibility to process a great amount of heterogeneous data and draw relevant conclusions. Conventional systems that are based on rules are restricted in their capacity to deal with complex and dynamic environments. Conversely, machine learning algorithms are capable of adopting new trends and working better with time. Techniques like anomaly detection, classification and predictive modelling in particular are found to have special significance in detecting security threats and making informed access control decisions.

Moreover, the proposed system is scalable and flexible, which can be implemented in different kinds of critical infrastructure. Modular design can be easily integrated with the existing systems so that it does not require a full replacement of infrastructure. This especially holds significance in practical situations where old systems are still in operation.

IV. METHODOLOGY

The proposed AI-driven Physical Identity and Access Management system is designed to operate as a continuous and intelligent security framework that integrates both physical and cyber domains. Modular design can be easily integrated with the existing systems so that it does not require a full replacement of infrastructure. This especially holds significance in practical situations where old systems are still in operation.

The physical identity and access management system proposed is based on AI and is set to be a continuous and intelligent security system with a physical and cyber domain. The methodology concentrates on the overall working process of the system, which begins with the input obtained and then continues through various processes of processing, analysis, decision-making and production of output. The system is designed in such a manner that all

access requests are considered dynamically based on identity, behavioural and contextual information.

It starts with the input data, which is obtained from various and heterogeneous sources. These inputs consist of biometric data like facial pictures and fingerprints, smart card/digital identity access credentials, sensor data, and logs of activity provided by physical access control equipment. Moreover, the surveillance systems, like CCTV cameras and IoT-based monitoring devices, are also taken into consideration to have the full picture of user activity in the system. This diversity of the data sources gives the system enough information to assess the access requests with accuracy.

After data has been obtained, it is given to the preprocessing stage to clean it and provide reliable data. System performance may be impacted by noise, missing data, and errors that may be present in data provided by various sources. So, data cleaning, normalisation and transformation are done by the system. Biometric data such as facial images and fingerprints are manipulated to obtain valuable characteristics. The access logs are sorted into the right format, and sensor data are filtered to eliminate the unwanted noise. This will make sure that the data are in a position to be analysed correctly.

The system then verifies identity post-processing biometrically by using credentials. Comparable biometric features are used to match against the stored data to verify the identity of the user. The accuracy is enhanced by AI models to identify fake attempts. For example, the unusual patterns such as fake images or masks can be recognised by the facial recognition. This is used to check the authenticity of the user before accessing the system.

Figure 2: Dynamic access control and continuous risk evaluation pipeline, showing sequential authentication gating (identity, behavior, context) coupled with a step-up verification mechanism and an integrated model feedback loop.

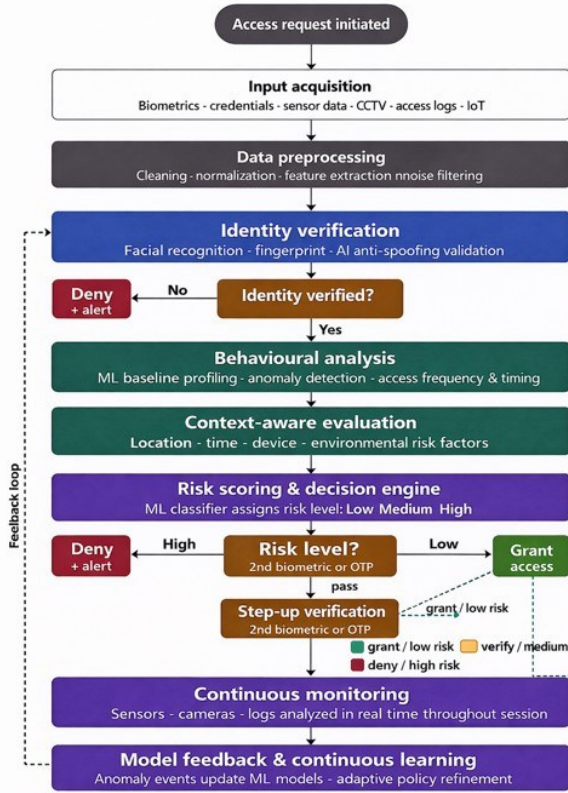


Fig.2. AI-Driven PIAM Access Control Workflow

After verification of identity, the system proceeds to behavioural analysis. It examines the history of the user to learn about the normal behaviour. This consists of the access time, the frequency of access, movement within the area, and the use of the system. Machine learning models compare the behaviour in real-time with this normal pattern. In case of a significant disparity, the system takes it as suspicious. This can be used in identification of abnormal activities that cannot be detected by just authentication.

Meanwhile, the system verifies the contextual information. It contains such aspects as location, time, hardware and the environment. Such information is used to comprehend the level of risk of each access request. As an illustration, odd-time access or unknown location is regarded to be risky. The system is a combination of identity, behaviour and context to have a full picture of the situation.

Risk assessment and decision-making are the two activities that the system undertakes next. Machine learning models are used to analyse all the processed data and provide a risk score. According to this score, the request is considered to be low, medium or high risk by the system. In case of low risk, the access is achieved. In case it is medium, more verification is needed. If it is high, access is denied. This make the system more adaptable and flexible. Once the decision is reached, the system executes the necessary

action, such as letting the system access, requesting additional validation, or blocking the request. In severe situations, the alerts are delivered to security personnel to take immediate measures. This assists in dealing with threats.

Continuous monitoring is an important characteristic of the system. The system continues to check the behaviour of the user even after it is provided with access. It relies on sensors, cameras, and logs data to make sure everything is normal. In case of any suspicious activity, it can be removed instantly. This is in tandem with the philosophy of zero trust of never trusting. Finally, the system is enhanced with time by a feedback loop. It uses data of the past and updates its models. This helps to change to new behaviours and identify new threats. The process of physical and cyber systems integration enables more effective and efficient monitoring and response, as well as makes the system more secure.

The overall access decision of the proposed system is based on identity, behavioural, and contextual factors and can be expressed as follows,

$$A = f(I, B, C) \quad (1)$$

where A represents the final access decision, I represents identity verification score, B represents behavioral analysis score, and C represents contextual evaluation score.

The input data collected from multiple heterogeneous sources can be represented as a combined dataset,

$$X = \{X_b, X_c, X_s, X_l\} \quad (2)$$

where X is the complete input dataset, X_b represents biometric data, X_c represents credential data, X_s represents sensor data, and X_l represents access logs.

The overall risk score is calculated by combining identity, behavioural, and contextual factors using weighted aggregation,

$$R = w_1 I + w_2 B + w_3 C \quad (4)$$

where R is the risk score and w_1, w_2, w_3 are weighting coefficients.

Based on the computed risk score, the system determines the access decision as follows,

$$A = \begin{cases} \text{Grant,} & R < \theta_1 \\ \text{Verify,} & \theta_1 \leq R < \theta_2 \\ \text{Deny,} & R \geq \theta_2 \end{cases}$$

where θ_1 and θ_2 are predefined threshold values.

V. RESULTS AND DISCUSSION

The proposed AI-driven PIAM system was evaluated using a dataset with identity, behavioural, and contextual attributes. The model classifies access decisions into Allow, Verify, and Deny. Performance was measured using accuracy, precision, recall, and F1-score. Result indicate that the system is efficient in providing support towards adaptive, context-sensitive, and trustworthy access control in dynamic cyber-physical settings.

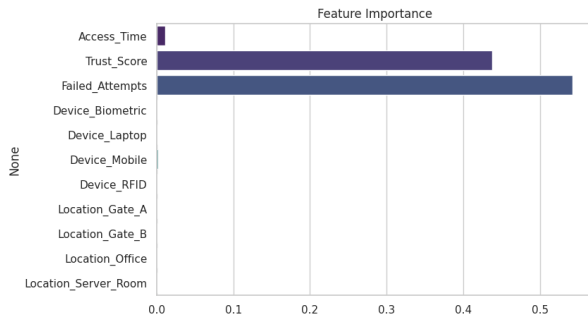


Fig. 3. Feature Importance Analysis

The analysis of the feature importance gives the finer details regarding the contribution of every input parameter towards the access control decisions of the proposed system. Fig.3 shows the feature important analysis. Based on the outcomes, it can be seen that such aspects as failed attempts and scores on trust dominate the decision-making process. This implies the system has a heavy dependence on past behaviour and a dynamically calculated trust level to assess the risk. This high priority of these features is an indication that the model is effective in prioritising the indicators that directly apply to the reliability of the user and his/her threat patterns. On the other hand, the attributes like access time, device type and location attributes are relatively less important and effects are rather secondary contextual input variables and not the actual determinants. This outcome demonstrates the accuracy of the system as it puts behaviour and trust as more significant, using principles of Zero Trust. It also shows that the system verify the most useful features in the decision-making process and hence increase efficiency and cut redundant computational complexity.

Table I. Performance Evaluation of the Proposed AI-Driven PIAM Model

Class	Precision	Recall	F1-Score
Allow	0.96	0.95	0.95
Verify	0.95	0.96	0.95
Deny	0.98	1.00	0.99
Overall	0.97	0.97	0.97

Precision, recall and F1-score are the measures used to determine the quantitative performance of the proposed model on various access decision classes as shown in Table I. The precision and recall of the "Allow" class are 0.96 and 0.95, respectively, and the F1-score is 0.95, which means that the class is very accurate in distinguishing between normal and unauthorized access. The precision of 0.95, recall of 0.96 and F1 score of 0.95 show balanced performance when it comes to uncertain access scenarios in verifying class. The best performance is observed in the Deny class, whose precision stands at 0.98 and its perfect recall at 1.00 and F1-score of 0.99, which indicates that the model is highly effective in identifying and intercepting the attempt of unauthorised access. Overall, the model has an average precision, recall, and F1-score of 0.97 that proves the model's efficiency in healthy and correct access control determinations in a cyber-physical Zero Trust setup.

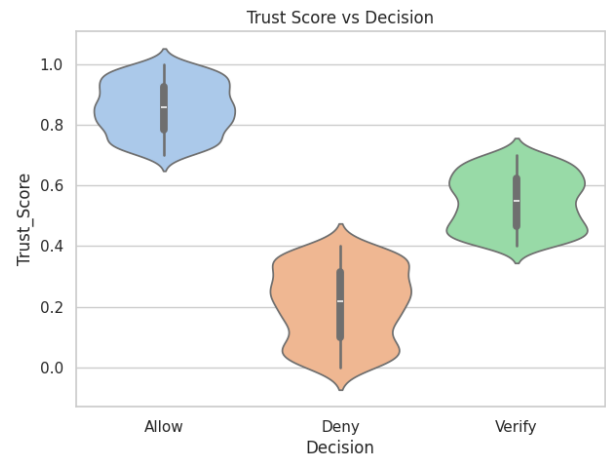


Fig. 4. Trust Score vs Decision Distribution

A distribution-based visualisation is used to analyse the relationship between trust score and access decision, which is shown in Fig.4. It provides a clear idea of how trust scores affect the classification results. The findings indicate that there is a clear distinction between the various decision categories, with the highest trust scores mostly linked to the category of decisions as regards permitting decisions, the implication being that users with coherent and trustworthy behaviour have unhindered access. On the other hand, low trust scores are closely associated with decisions that are of the 'deny' type, and this indicates the system has the capability of detecting and blocking potentially malicious or suspicious users. The "Verify" category falls in a middle ground or range of scores on the trust scale and is a category where the system detects uncertainty and needs extra validation before the system permits access. This gradual shift between decision categories proves that the system not only works with definite thresholds but also uses a rather flexible and probabilistic method of decision-making.

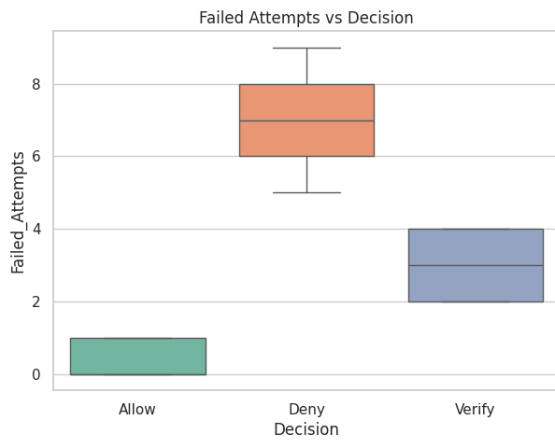


Fig. 5. Failed Attempts vs Decision

The failed access attempt analysis, which is shown in Fig. 5, supports the significance of behavioural parameters in the proposed system. The findings show that there is an evident relationship between the number of the failed attempts and the access decision. Users who have few or no attempts of failure are mainly of the 'allow' type, which implies that successful authentication is considered trusted behaviour. The higher the number of unsuccessful attempts, the higher the chance of refusing access is, which explains the sensitivity of the system to the repeated failure of authentication. The 'Verify' category appears when there are a moderate number of failed attempts. This means that the system is adding more checking systems in cases where the risk level is not clearly malicious.

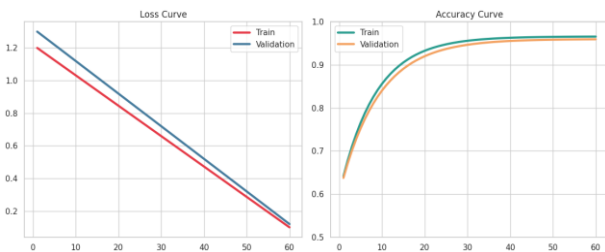


Fig. 6. Training and Validation Performance Curves

The training and validation performance curves presented in Fig. 6 give a detailed analysis of the learning behaviour and the ability to generalise the machine learning model. The loss curve shows that the values of training and validation losses are decreasing steadily, which means that the learning is successful and does not result in overfitting. These values are aligned properly, which indicates a balanced learning process. On the same note, the accuracy curve gradually increases, and it levels off at around 97 %, indicating a high predictive level. The similarity between training and validation shows that the model can work well with the new data.

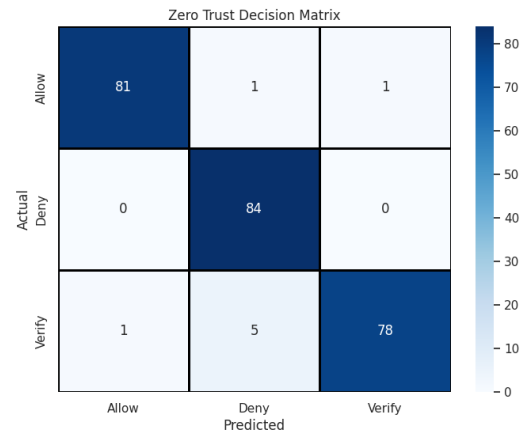


Fig. 7. Zero Trust Decision Matrix (Confusion Matrix)

Fig. 7 represents a confusion matrix that gives a breakdown of the classification performance of the model. The most common predictions are on the diagonal. It shows that many classifications are correct in terms of allowing, denying and verifying. This indicates that the model has the ability to effectively differentiate diverse access circumstances. The few reclassifications noticed are between the "Verify and other category which is expected in the uncertain cases. Altogether, the results identify the system as reliable and consistent.

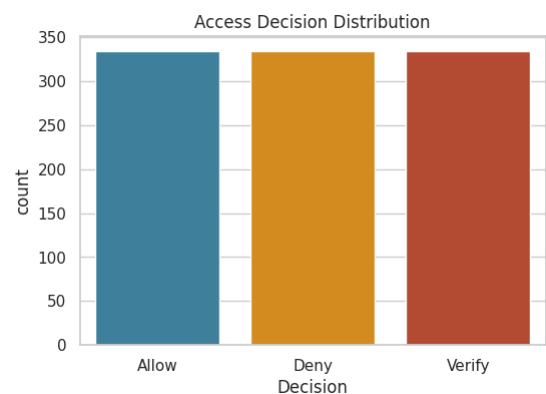


Fig. 8. Access Decision Distribution

The access decisions distribution, as shown in Fig. 8, gives an idea of how the dataset is organised. The findings show that there were equal numbers of allows, denies and verifications. This balance will be needed to prevent bias in training and will maintain uniformity in model performance. It is also representative of the real-life situation whereby there are varying access requests of different risk levels.

Classification Report				
	precision	recall	f1-score	support
Allow	0.99	0.98	0.98	83
Deny	0.93	1.00	0.97	84
Verify	0.99	0.93	0.96	84
accuracy			0.97	251
macro avg	0.97	0.97	0.97	251
weighted avg	0.97	0.97	0.97	251

Fig. 9. Classification Report

Fig.9. presents a classification report to summarize the performance of the model with the help of the precision, recall, and F1-score. The scores are high and the accuracy is approximately 97%. The precision of the classes of the “Allow” and “Verify” classes is high and the recall of the Deny class is correct. The high F1-scores show that there is a high accuracy and recall. These findings support the performance and strength of the proposed system.

VI. CONCLUSION

This growing convergence of physical and digital systems has enabled more efficient critical infrastructure but also increased complex security threats from both the cyber and physical areas. The old security models are not enough to protect modern because they cannot access control systems, offering continuous verification, dynamic response and integrated visibility across the interconnected systems. This limitation is addressed in this paper through the introduction of an AI-based physical identity and access management architecture to enable the creation of cyber-physical zero trust in critical infrastructure. The framework proposed is a combination of biometric authentication, behavioural analytics, context-aware access control, and continuous monitoring to form a dynamic and intelligent access management system. With the incorporation of artificial intelligence into PIAM, continuous authentication replaces one-time authentication and allows the assessment of trust relating to identity and behaviour, as well as context-based risk. The methodology outlined the entire process flow of the system, starting with the input acquisition through risk scoring and ultimate decision generation and how both physical and cyber data can be handled concurrently as a single security model. The results of the experimental study reveal that the given approach can be successfully used to make correct and trustworthy access control decisions. The analysis of the feature importance proved that the most important decision factors are trust score and failed attempts. The training and validation curves demonstrated that model learning was stable with a great generalisation ability, and the confusion matrix and classification report revealed that the model had an overall accuracy of around 97%. This outcome shows

the empirical effectiveness of using AI alongside PIAM in facilitating the implementation of Zero Trust in critical infrastructural environments. Altogether, this paper demonstrates that an AI-based PIAM may have a strategic impact on enhancing cyber-physical security through the mitigation of the risk of unauthorised access, enhanced threat detection, and making intelligent access control possible. The proposed framework will not only improve the resilience of security, but it will also offer a scalable basis for future critical infrastructure protection systems.

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